

Guided Activity

Scenario: You have been asked to analyse and visualise the monthly sales data of a company for the year 2023.

Using the data file provided by your trainer, complete the following tasks. This activity will assume using Excel; however, you may use whatever application you are familiar with.

1. Open the file **QuantitativeData.xlsx** and select the **Raw Data** sheet.
2. Starting in cell D1, input a label - **Total Sales**
3. In cell D2, input a label – **Average Sales**
4. In cell E1, create a function to calculate the total sales for the year. You can use the *SUM* function to add up all the sales amounts.
5. In cell E2, Calculate the average monthly sales for the year. You can use the *AVERAGE* function to find the mean of the sales amounts.
6. In cell D4, input a label – **Highest Sales**
7. In cell D5, input a label – **Lowest Sales**
8. In cell E4, create a function to calculate the MAXIMUM sales for the year. You can use the *MAX* function.
9. In cell E5, Calculate the LOWEST monthly sales for the year. You can use the *MIN* function.
10. Create a conditional format rule that highlights the MAXIMUM sales for the year. You can use the following formula to achieve this
=B2=MAX(\$B\$2:\$B\$13)
11. Create a **column chart** to visualise the monthly sales data. The months should be on the x-axis and the sales amounts on the y-axis.

Review activity - Analysing sales data (using Excel)

Objective: In this activity, you will learn to perform simple quantitative analysis using Microsoft Excel.

You will analyse sales data to gain insights into revenue trends, best-selling products, and sales performance.

Analyse the data:

1. Open the file called QuantitativeData.xlsx
2. Select the sheet tab called **Sales Data**
3. In cell E1, create a label **Gross Revenue**
4. In cell E2, calculate the gross revenue for the first sale. You could use the formula “=C2*D3”
5. Drag the formula down to calculate the total revenue for all sales.
6. In cell G1, create a label **Total Units Sold**
7. In cell H1, calculate the total units sold. You could use the formula “=SUM(C2:C13)”

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8. In cell G2, create a label **Best-Selling Product**
1. In cell H2, find the best-selling product. You could use the formula `"=INDEX(B:B,MATCH(MAX(C:C),C:C,0))"`
2. In cell G4, create a label **Average Units Sold**
3. In cell H4, calculate the average units sold. You could use the formula `"=AVERAGE(C2:C13)"`
4. In cell G5, create a label **Total Gross Revenue**
5. In cell H5, calculate the total gross revenue. You could use the formula `"=SUM(E2:E13)"`

Creating a Line Chart:

1. Continuing with the same data as before.
2. Select the range of data (including the headings) from columns A, C, and E.
3. Go to the *"Insert"* tab and choose an appropriate *"Line Chart"* to create a chart representing sales over time.

Formatting and Final Touches:

1. Apply appropriate formatting to the cells, such as bold headings, data alignment, and currency formatting.
2. Add a title, label axes, and format the chart as needed.
3. Update the changes to the original workbook.

If time allows:

- Calculate the Net Revenue by using the following formula:

Net Revenue = Gross Revenue – Gross Revenue * 20%

- Create a pivot table to summarise sales data by product and date.
- Use conditional formatting to highlight cells with matching the following target:

Net Revenue >= £100 – these cells should go GREEN